

AREE 2.0

Air Pollution–Weather Coupled Forecasting System for Delhi NCR

Technical architecture, module specification, algorithm selection and team allocation. Includes the validated **aerosol–boundary-layer feedback loop gain (λ)** that is the scientific core of this submission, its evidence base, and the runnable pipeline that proves or kills it before any model is built.

WHAT THIS DOCUMENT DECIDES

- **The scientific differentiator** — λ , the loop gain of the aerosol–radiation–PBL feedback, and the moment it crosses into self-amplification.
- **What is reused from the existing AREE codebase** — audited file by file, not assumed.
- **The forecasting algorithm** — and why every alternative was rejected.
- **The module boundaries** — five owners, minimal coupling, each independently testable.
- **The evaluation protocol** — the numbers NCMRWF will ask for.

STATUS AT TIME OF WRITING

The pipeline in `aree-ps26082/scripts/` is written and passes its synthetic self-test: the estimator recovers a prescribed λ of 0.1386 as 0.1513, a 9.2% error, with all three physical sign checks passing. It has **not** yet been run on real ERA5 data — that requires a Copernicus CDS key and is step one of the work plan in section 11.

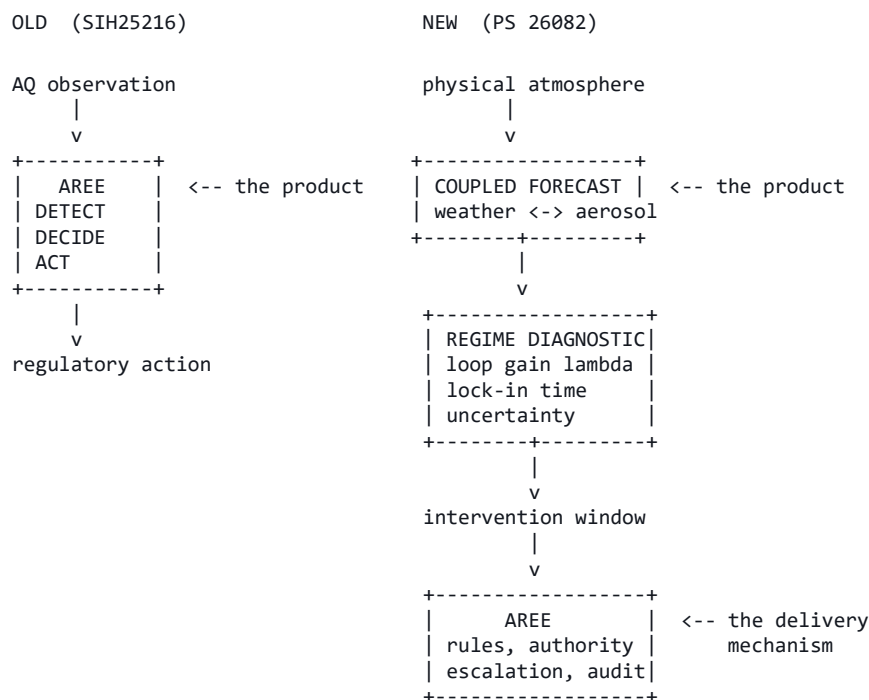
1 The correction this architecture makes

PS 26082 is owned by NCMRWF — India's operational numerical weather prediction centre — and sits under the **Disaster Management** theme. Both facts change what the deliverable is.

The previous AREE positioning was built for SIH25216, a policy-dashboard problem where the winning wedge was governance: everyone else stopped at a recommendation, so AREE owned the regulatory execution layer after it. That wedge does not transfer. Here the evaluators are atmospheric scientists and the deliverable they are judging is forecast skill under coupling. Leading with escalation workflow would read as last year's project brought to a different problem.

The inversion

AREE does not die. It changes position in the stack. It stops being the product and becomes the operational layer of a forecasting system — what converts a physical prediction into lead time someone can act on, which is precisely what Disaster Management means.



The one-line positioning

Everyone forecasts the pollution level. We forecast the moment the atmosphere stops responding to intervention.

DO NOT USE THIS LINE YET

It is only true if section 4's validation returns a usable separation on real data. If it does not, the honest fallback is a coupled forecasting system with an explicit feedback diagnostic and no lock-in claim. Find that out privately, in week one, not in front of judges.

2 Audit of the existing AREE codebase

Findings from github.com/jeyamoorthi/urbanlive-ai at commit e876f6d, cloned and read in full. This section exists because the reuse plan has to be based on what the code does, not on what the pitch deck says it does.

What is genuinely there and worth keeping

COMPONENT	FILE	ASSESSMENT
Pathway streaming DAG	backend/app.py	Real. pw.io.python.read with custom ConnectorSubject, genuine pw.temporal.sliding windows at 3/5/15 min with 1 min hop, pw.reducers aggregation. Not a timer loop dressed as streaming.
GRAP state machine	streaming/state_machine.py	Clean deterministic implementation. Hysteresis requires HYSTERESIS_CONFIRMATIONS consecutive agreeing windows before a stage transition. Per-station state. Directly reusable.
Persistence tracker	streaming/state_machine.py	Counts consecutive high-AQI windows before triggering. Correct pattern, wrong timescale for this PS — see below.
Causal attribution	streaming/risk_engine.py	CausalAttributionEngine + TransportVectorModel: FIRMS fire counts joined with wind bearing. Reusable as an emission-proxy input.
FIRMS ingestion	ingestion/firms_stream.py	Working NRT satellite fire pull with confidence filtering and bbox query. Keep as-is.
Regulatory RAG	rag/advisory_engine.py	Document-grounded advisory generation over the policies directory. Keep, demote in prominence.
Audit and reporting	report_generator.py	PDF brief generation. Keep as the order-export path.
Next.js operations UI	frontend/	Next 16, React 19, Recharts, Leaflet. Real dashboard. The spatial forecast layer plugs into the existing Leaflet map.

What is missing, and it is more than the pitch assumes

CRITICAL FINDING

backend/ingestion/weather_stream.py is **zero bytes**. The current system ingests no meteorology at all. Temperature, humidity, pressure, boundary layer height and radiation are absent from the codebase entirely. For a problem statement whose entire subject is weather–chemistry coupling, this is the gap that defines the work.

GAP	CURRENT STATE	REQUIRED BY PS 26082
Meteorology	weather_stream.py is empty (0 lines). Only wind speed and bearing arrive, embedded in the WAQI payload.	Full met state: T, RH, wind, pressure, PBL height, shortwave down, clear-sky shortwave.
Forecast method	compute_short_term_forecast() in app.py: numpy.polyfit degree 1 on recent AQI, extrapolated linearly.	Coupled physical evolution over 72 hours.

Forecast horizon	5 and 30 minutes.	72 hours — a 144x extension. Linear extrapolation is not a candidate at this horizon.
Spatial structure	5 point stations, only 2 inside Delhi NCR.	Gridded NCR domain.
Coupling	None. Chemistry and meteorology never interact.	Two-way aerosol–radiation–PBL feedback — the explicit subject of the PS.
Persistence timescale	PERSISTENCE_THRESHOLD = 3 windows of 3 minutes.	Meteorological episodes persist over 12–72 hours. The mechanism is right, the constant is three orders of magnitude too small.
Verification	None in the repository.	MAE, RMSE, bias against persistence and an external baseline.

Two claims to stop making immediately

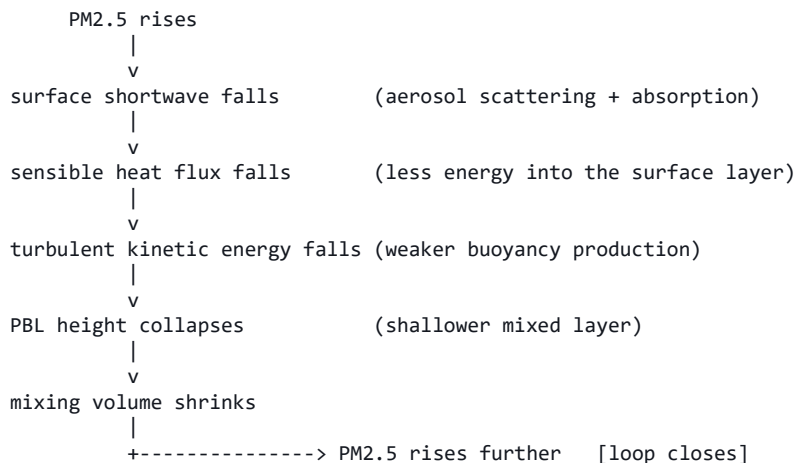
- **The 23 ms figure.** That is React/API round-trip latency. It has nothing to do with forecasting. Quoted anywhere near a 72-hour physical forecast, an NWP scientist reads it as a category error and the room is lost. Report it, if at all, as “decision-layer response latency” in a separate section.
- **“Streams every thirty seconds.”** AQI_POLL_INTERVAL = 30 is the poll interval. WAQI republishes roughly hourly, which the codebase already acknowledges — FRESH_DATA_THRESHOLD_SECONDS is set to 90 minutes. Polling frequency is not measurement freshness. Report observed_at, received_at and processed_at separately and display data age. That is more credible, not less.

3 The scientific core: the feedback loop gain λ

3.1 The mechanism, in the problem statement's own words

The PS says: “dense concentrations of aerosols (PM2.5) block sunlight, altering local temperatures, wind patterns, and planetary boundary layer (PBL) heights... Ignoring these coupled meteorological-chemical feedback loops leads to significant inaccuracies.”

That sentence describes a positive feedback. Written out:



Every other team will treat this coupling as an implementation detail on the way to a better AQI number. Their output is a line graph. The differentiator proposed here is to treat the loop as a diagnosable state with a critical threshold, and to report the threshold crossing as the primary product.

3.2 Formulation

Take the three log-sensitivities (elasticities) around the ring:

$e1 = d \ln C / d \ln H$ PM2.5 response to boundary-layer depth
 $e2 = d \ln H / d \ln S$ PBL response to surface shortwave
 $e3 = d \ln S / d \ln C$ shortwave response to aerosol load

$\lambda = e1 * e2 * e3$

Expected signs follow from elementary physics:

TERM	SIGN	PHYSICAL BASIS	EXPECTED MAGNITUDE
e1	negative	Box dilution. For fixed emission E, $C \sim E / (H \cdot U)$, so $d \ln C / d \ln H = -1$ exactly.	≈ -1
e2	positive	Encroachment growth. Mixed-layer depth scales with the time-integrated surface heat flux, $H \sim S^{1/2}$.	$\approx +0.5$
e3	negative	Beer–Lambert extinction. Optical depth rises with column aerosol mass, transmitted flux falls exponentially.	-0.1 to -0.4

The product of $(-)(+)(-)$ is **positive**. A positive loop gain is a self-reinforcing feedback. The closed-loop amplification of any perturbation is then

$$A = 1 / (1 - \lambda) \quad \text{for } \lambda < 1$$

which is the standard linear feedback-factor form used throughout geophysics — the same algebra as the climate feedback parameter (Roe, **Feedbacks, Timescales, and Seeing Red**, Annual Review of Earth and Planetary Sciences 37:93–115, 2009). As λ approaches 1 the amplification diverges: the atmosphere stops damping perturbations and the episode locks in.

WHY ELASTICITIES AND NOT RAW PARTIAL DERIVATIVES

Elasticities are dimensionless, so λ is dimensionless, comparable across cities, seasons and years, and independent of the units anyone reports PM2.5 or PBL height in. A product of raw partial derivatives would carry units of $(\mu\text{g m}^{-3} / \text{m})(\text{m} / \text{W m}^{-2})(\text{W m}^{-2} / \mu\text{g m}^{-3})$, which happens to cancel — but only by construction, and the intermediate quantities would not be interpretable on their own.

3.3 Evidence base

The mechanism is established in the peer-reviewed literature. What is novel here is operationalising it as a live, forecastable scalar with a critical threshold — not the physics itself. That distinction must be stated explicitly in the presentation; claiming λ as a published standard index would not survive a single question.

CLAIM IN THE ARCHITECTURE	SUPPORTING LITERATURE	WHAT IT CONSTRAINS
Aerosols measurably dim the surface over Delhi / the Indo-Gangetic Plain during haze episodes.	WRF-Chem studies of aerosol optical properties and surface shortwave over Delhi report reductions of order 25–80 W m ⁻² during polluted episodes.	Magnitude of e3.
Reduced surface heating suppresses turbulence and collapses the mixed layer.	Winter Fog Experiment (WiFEX) and IGP haze studies report daytime maximum PBL height falling from roughly 1.3 km in clean conditions toward roughly 0.6 km under heavy aerosol loading.	Magnitude of e2 and the operating range of H.
Absorbing aerosol aloft stabilises the column — the “dome effect”.	Ding et al., Geophysical Research Letters, 2016, on black-carbon induced boundary-layer suppression; Petäjä et al., Scientific Reports, 2016, on aerosol–boundary-layer feedback in polluted megacities.	Existence and sign of the loop.
Compressed, humid boundary layers accelerate secondary aerosol formation, strengthening the loop.	Studies of the secondary-aerosol positive feedback during severe haze over the Beijing–Tianjin–Hebei region; Delhi speciation work showing organics at roughly 50–69% of non-refractory submicron mass with a large chloride fraction from solid-fuel combustion.	Why e3 must not be estimated with RH controlled out.
Feedback factors compose as $A = 1/(1-\lambda)$.	Roe, Annual Review of Earth and Planetary Sciences, 2009 — the canonical treatment of feedback composition in geophysical systems.	The algebra of the threshold claim.
Ventilation coefficient (PBLH × wind) is the conventional dispersion metric.	Standard in operational air-quality meteorology.	The baseline λ must outperform.

VERIFY BEFORE SUBMISSION

Every citation above should be pulled, read and page-checked by the team before it appears on a slide. They are given here as the correct places to look, from a literature survey — not as verified page references. A wrong citation in front of NCMRWF costs more than no citation.

3.4 The trap the estimator is built to survive

PM2.5, PBL height and surface shortwave all have large, in-phase diurnal cycles. A naive regression of $\ln C$ on $\ln H$ recovers the diurnal cycle, not the feedback — and returns a beautiful, meaningless $e1$ near -1 every single time. This is the single most likely way for this project to produce a confident wrong answer.

The fix, applied identically to all four variables in `log_anomaly()`: every variable is converted to a log anomaly from its own (month, hour-of-day) climatology before any regression. What remains is the perturbation about normal diurnal evolution — exactly the quantity a feedback gain is defined on.

3.5 Self-test result

The estimator was tested against synthetic data with a prescribed loop gain, generated with deliberately large diurnal cycles so that a de-seasonalisation failure would show up as a test failure rather than passing quietly.

```
synthetic ground truth:  e1=-0.90  e2=+0.55  e3=-0.28
                        TRUE lambda = 0.1386
```

```
physical sign checks
```

```
-----
PASS  e1 = dlnC/dlnH  should be NEGATIVE (dilution)      -1.961
PASS  e2 = dlnH/dlnS  should be POSITIVE (radiative)      +0.687
PASS  e3 = dlnS/dlnC  should be NEGATIVE (attenuation)    -0.160
```

```
recovered lambda  +0.1513  (median of rolling estimate)
relative error     9.2%
PASS - estimator recovers the prescribed loop gain
```

```
separation on synthetic episodes
```

```
lambda          AUC 0.733  median lead time 12 h
negative ventilation coeff.  AUC 0.433  median lead time 22 h
```

KNOWN LIMITATION – REPORT THIS, DO NOT HIDE IT

The individual elasticities are biased: $e1$ came back at -1.96 against a true -0.90 . This is **simultaneity bias**, the classic endogeneity problem when ordinary least squares is applied to a closed loop — each regressor is correlated with the error term because the system is solved jointly. The product λ survives because the biases partially cancel, but this is luck, not design. The correct fix is instrumental-variable or simultaneous-equations estimation, using an instrument that shifts one variable without acting through the others — synoptic wind is the natural candidate. This is a real piece of work and it belongs to Module B.

4 Validation protocol: the go / no-go gate

The central claim is falsifiable and must be falsified or confirmed before anything is built on it. The pipeline does this without a forecast model, using observations alone, which also means the result survives even if the emulator underdelivers.

The circularity that had to be designed out

WHY λ IS COMPUTED UPSTREAM, NOT DOWNSTREAM

In the earlier architecture sketch λ was computed from the emulator's forecast fields. That is circular: if the emulator predicts PM2.5 rising and PBL collapsing, λ mechanically comes out high, and the diagnostic is just a re-reading of the model's own output. The first question a reviewer asks is whether λ says anything the PM2.5 curve did not. Computing λ from observations alone, over past winters, before any model exists, removes both the circularity and the single point of failure.

The test

STEP	WHAT HAPPENS	GUARD AGAINST
Label episodes	Contiguous runs above a PM2.5 threshold, short gaps bridged. "Locked in" requires BOTH duration ≥ 48 h AND peak $\geq 250 \mu\text{g m}^{-3}$.	Labels never see λ . A long mild event is not a lock-in; a brief sharp local spike is not either.
Read the predictor	Maximum λ in the 24 h BEFORE onset.	Reading before onset is what makes it a forecast test rather than a description.
Separation	ROC AUC via the Mann–Whitney identity, hand-implemented.	No sklearn dependency, no hidden weighting; the computation is auditable line by line.
Threshold sweep	Full confusion table across all candidate thresholds; Youden J reported but not imposed.	The operating point is a policy judgement about the cost of a false GRAP invocation, not a modelling choice.
Lead time	Hours between first threshold crossing and onset.	A diagnostic firing 2 h ahead is a description. One firing 18 h ahead is decision support.
Baseline	The identical test using negative ventilation coefficient.	If λ cannot beat PBLH $\times$ wind, it adds nothing and should be dropped.
Holdout	November 2023 and November 2024 excluded from every fit.	The only numbers quoted publicly come from episodes the fit never saw.

Decision rule

HELD-OUT AUC	VERDICT	ACTION
≥ 0.75	λ separates the classes.	Proceed. λ becomes the headline and the emulator is built to forecast it.
0.60 – 0.75	Weak separation.	Fix the estimator first — instrumental variables for the simultaneity bias, ceilometer-corrected PBLH. Do not present the lock-in claim yet.

< **0.60**

λ does not separate the classes.

Drop the lock-in claim entirely. Fall back to a coupled forecasting system with an explicit feedback diagnostic — still responsive to the PS, still ahead of an uncoupled ML forecaster.

5 Algorithm selection

5.1 The forecasting engine

The PS names WRF-Chem, and its own wording gives the escape hatch: “WRF-Chem or similar open-source coupled frameworks”. That phrasing is deliberate and should be taken.

REJECTED: WRF-CHEM IN THE OPERATIONAL LOOP

- Requires an emissions inventory, chemical boundary conditions, a chemical mechanism and hours of HPC time per forecast cycle.
- Cannot be demonstrated live. Every team that promises this will be showing pre-computed output, a toy domain, or nothing.
- A five-person student team cannot operate it on a hackathon timeline.

REJECTED: PURE SEQUENCE MODEL ON AQI HISTORY

- An LSTM or Transformer trained on historical AQI has no causal handle on emissions. It cannot answer “what if construction halts at T+6” — the counterfactual is structurally outside its hypothesis space.
- It does not represent the coupling the PS is explicitly about, so it fails the brief even if its error metrics are respectable.
- It is what most competing teams will build.

SELECTED: PHYSICS-CONSTRAINED COUPLED EMULATOR

A neural emulator of a coupled reference model, with the feedback terms carried as explicit, inspectable state rather than hidden weights.

DESIGN CHOICE	WHAT IT IS	WHY THIS AND NOT THE ALTERNATIVE
Backbone	Convolutional encoder–decoder (U-Net shape) over the NCR grid, with a recurrent step advancing one hour at a time for 72 steps.	The state is a spatial field on a regular grid and the physics is local advection and diffusion — exactly what convolution encodes. A graph network buys nothing on a regular grid; a plain MLP throws away spatial structure. Autoregressive hourly stepping matches how the physical model itself integrates, so the emulator learns a transition operator rather than a 72-hour black box.
Explicit feedback channels	PBL height, surface shortwave and aerosol load are separate output channels, not internal activations.	This is what makes λ computable from the forecast at all. A model that only emits AQI cannot be interrogated for regime state, and the entire differentiator disappears.
Monotonicity constraint	Sign of the aerosol→radiation and radiation→PBL responses constrained during training via a penalty term.	Prevents the emulator learning a physically inverted feedback that happens to fit the training set. This is what “physics-informed” means concretely, rather than as a slogan.
Loss	Weighted MSE in log space, plus a term penalising error in the timing of threshold crossings.	Log space because PM2.5 is log-normally distributed and plain MSE would be dominated by the few extreme hours. The timing term exists because for Disaster Management, being right about WHEN matters as much as being right about HOW MUCH.

Uncertainty	Small ensemble from perturbed initial conditions plus MC dropout; spread reported alongside every forecast.	Cheaper than a full ensemble system, and it lets the system report when not to trust itself — which for a disaster-management product is worth more than a confident wrong number.
Training reference	A coupled reference dataset — WRF-Chem runs over historical Delhi episodes, and/or open coupled products (CAM5, SILAM, GEOS-CF) as both boundary conditions and the baseline to beat.	Deliberately not fixed yet. Committing to a training source before confirming the data can actually be obtained is how projects die in week five. Module B settles it in week two.

5.2 Why every module is written as functions

This is not a style preference. Three specific properties are being bought:

- **Testability of the scientific claim.** `log_anomaly()`, `ols_slope()`, `auc_mann_whitney()` and `find_runs()` each have one job and a checkable output. An off-by-one in `find_runs()` silently shifts every episode boundary by an hour and corrupts the lead-time result — as a function it can be tested against hand-built cases; inline in a loop it cannot.
- **Substitutability at the boundaries.** The training reference is undecided and the PBL source may change from ERA5 to ceilometer-corrected. Because loading, deriving and estimating are separate functions, swapping the source touches one function, not the pipeline.
- **Auditability by a reviewer.** A judge who asks “how did you estimate that derivative” can be shown a fifteen-line function with no hidden regularisation. That is why ordinary least squares is hand-written rather than imported — there is visibly nothing else changing the answer.

5.3 Why Pathway

Pathway is already load-bearing in the existing codebase, and it earns its place here for a reason that is specific to this problem rather than generic streaming enthusiasm.

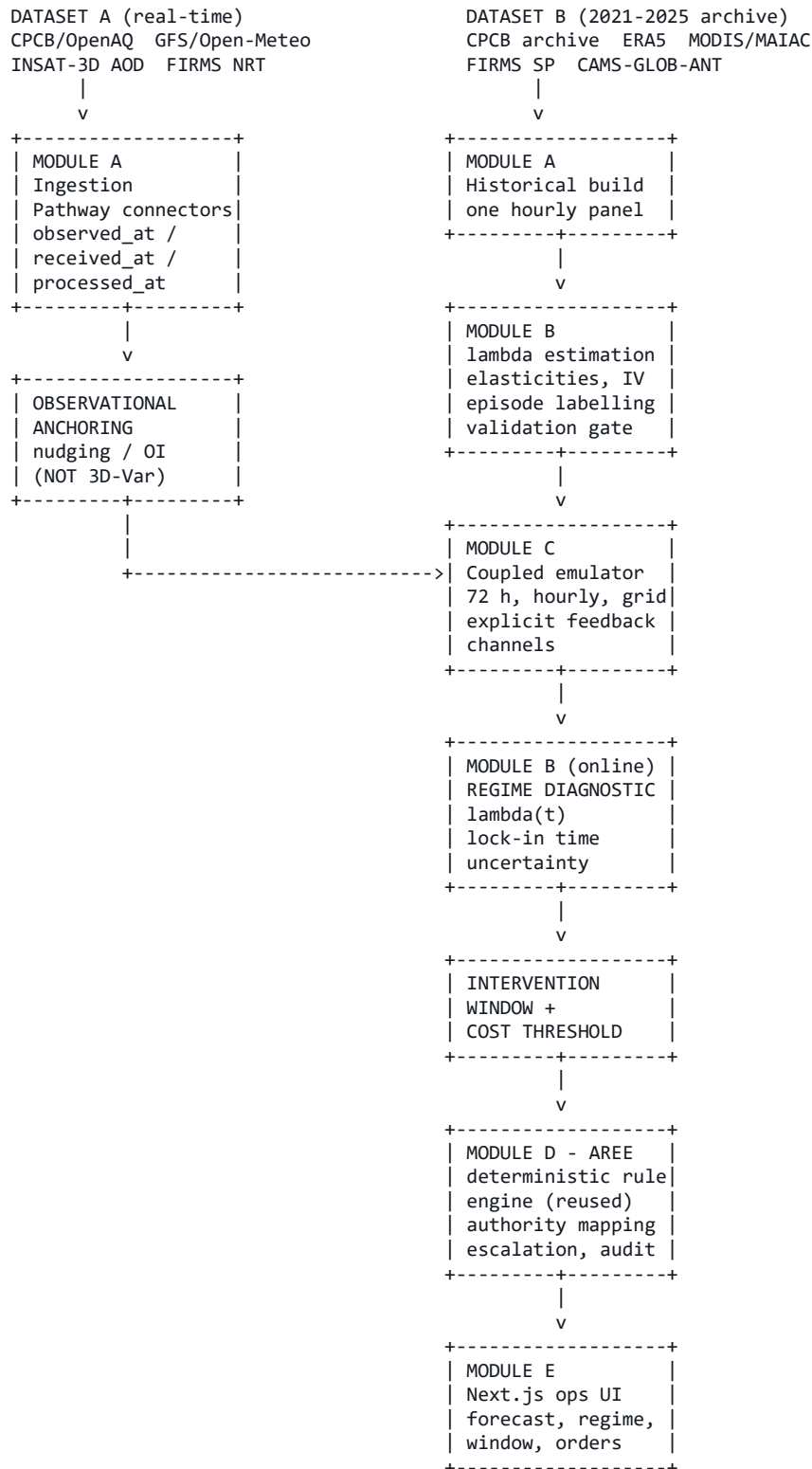
PROPERTY	WHY IT MATTERS FOR PS 26082
Event-time windows with late and out-of-order data	CPCB stations drop out and backfill. Satellite AOD arrives 30 minutes to 3 hours after overpass. FIRMS NRT lands within 3 hours. A processing-time system would either wait for the slowest source or silently compute on incomplete windows. Event-time semantics let each source arrive when it arrives and still land in the correct hour.
Incremental recomputation	When a late observation arrives, only the affected downstream state is recomputed rather than the whole window. This is what makes continuous re-anchoring of a 72-hour forecast affordable.
Replay determinism	The same input sequence produces the same output sequence. This is the foundation of the audit trail: an escalation decision can be replayed line by line months later. For a regulatory system, this is not a convenience, it is the product.
Continuous re-anchoring — the strategic point	NCMRWF runs operational cycles on a fixed HPC schedule. A streaming system can re-anchor its forecast against new observations far more often. This is the one axis on which a student team can legitimately outperform the institution that wrote the problem statement — not model physics, but update cadence and delivery.

THE HONEST CAVEAT

Pathway is the right tool for ingestion, windowing, assimilation triggering and the decision layer. It is not a numerical model runner. The emulator is invoked from the DAG; it does not live inside it.

6 Target architecture

===== AREE 2.0 =====



The layer that was missing from the previous sketch

FALSE-ALARM COST

GRAP Stage III has real economic cost — construction halts, school closures. So the question “at what forecast probability do you actually act?” is a decision-threshold problem with asymmetric costs, and it was absent from the earlier architecture. It belongs between the intervention window and the AREE core, and it is the best possible home for the deterministic engine: the threshold is encoded as policy, not decided by a model, and the audit trail records which threshold was in force when an order fired. The “AI explains, AI does not decide” principle extends into it cleanly.

7 Module specification and team allocation

Five owners. Module boundaries are drawn so that each can be developed and tested without waiting on another, and so that the interface between any two is a file on disk or a typed dictionary — never shared mutable state.

MODULE A — Data engineering

Mission. Own every byte that enters the system, and the honesty of its timestamps.

DELIVERABLES

- ERA5 pull for the NCR domain, 2021–2025, Oct–Feb (scripts/01_fetch_era5.py — written, needs a CDS key).
- Ground PM2.5 via OpenAQ v3 or the S3 archive (scripts/02).
- MODIS MCD19A2 AOD via Google Earth Engine; INSAT-3D AOD via MOSDAC.
- FIRMS NRT + SP ingestion — extend the existing firms_stream.py.
- Fill in the empty weather_stream.py with a real met connector.
- Three-timestamp discipline everywhere: observed_at, received_at, processed_at, and a data-age field surfaced to the UI.

INTERFACE OUT

data/processed/panel_hourly.parquet, and live Pathway tables with the same column names.

DEFINITION OF DONE

The panel builds end-to-end, ERA5 accumulations are correctly converted to W m^{-2} , and no step in the pipeline can tell whether it is reading live or replayed data.

MODULE B — Feedback science

Mission. Own λ : its estimation, its validation, and the honesty of the claim.

DELIVERABLES

- Run the validation gate on real data and report the held-out AUC (scripts/04–06 — written and self-tested).
- Fix the simultaneity bias with instrumental-variable estimation; synoptic wind is the natural instrument.
- Quantify ERA5 PBLH bias against IMD radiosonde over Delhi, and against ceilometer if a series can be obtained.
- Define the regime classification bands and the lock-in time estimator.
- Uncertainty conditioned on λ : does forecast spread widen as λ approaches 1?

INTERFACE OUT

validation_report.json, the regime thresholds, and a written statement of what λ does and does not support.

DEFINITION OF DONE

A held-out AUC number exists and is defensible, or the claim is formally withdrawn and the fallback positioning adopted.

MODULE C — Forecast emulator

Mission. Own the 72-hour coupled prediction.

DELIVERABLES

- Settle the training reference: WRF-Chem runs, open coupled products, or a combination. Decide in week two, not week five.
- Build the U-Net + recurrent-step emulator with explicit PBL, shortwave and aerosol output channels.
- Monotonicity penalty on the feedback terms.
- Ensemble spread from perturbed initial conditions.
- Verification: MAE, RMSE, bias at 24/48/72 h against persistence and an external baseline, on the held-out Novembers.
- Natural-experiment validation: the 2020 lockdown emissions perturbation and the annual Diwali spike, rather than synthetic counterfactuals.

INTERFACE OUT

A callable forecast(state) -> 72h gridded fields with uncertainty.

DEFINITION OF DONE

It beats persistence at 24 h, and where it does not, that is stated plainly with the horizon at which skill is lost.

MODULE D — Regulatory engine

Mission. Own the conversion of physical risk into accountable action.

DELIVERABLES

- Reuse state_machine.py essentially as-is — it is already correct.
- Retune PERSISTENCE_THRESHOLD from 3 windows of 3 minutes to a meteorologically meaningful horizon.
- Add the predicted-breach path alongside the observed-breach path.
- Build the false-alarm cost threshold layer.
- Jurisdiction → authority mapping; escalation case with deadline.
- Hash-chained audit; signed order export via report_generator.py.

INTERFACE OUT

Escalation cases with rule citation, authority, deadline and a replayable decision record.

DEFINITION OF DONE

Any escalation can be replayed from the audit log and reproduces byte-identically.

MODULE E — Interface and integration

Mission. Own what a human actually sees, and the seams between modules.

DELIVERABLES

- Forecast view: 72 h, gridded, on the existing Leaflet map.
- Regime view: λ through time with the threshold and the projected lock-in hour.

- Intervention window as the primary call to action, not a number.
- Data-age and provenance surfaced on every panel.
- Keep Next.js free of business logic — the existing discipline is correct and must hold.
- Docker compose, environment handling, end-to-end demo script.

INTERFACE OUT

A running system and a rehearsed demo path.

DEFINITION OF DONE

The demo runs twice in a row without manual intervention.

The fifth seat: presentation

The PPT owner is not a documentation clerk. They own the argument: the one-line positioning, the lock-in demo narrative, the verification slide, and the answers to the three questions NCMRWF will certainly ask — what is your skill score, how did you handle PBLH uncertainty, and what does a false alarm cost. That work starts now, not in the last week, because it determines which results the technical modules must produce.

8 Evaluation protocol

A forecasting problem statement judged by a forecasting centre will be asked for skill scores. A team with honest verification beats a team with a prettier dashboard in front of these judges, every time.

CATEGORY	METRIC	COMPARED AGAINST	REPORTED AT
Forecast skill	MAE, RMSE, mean bias on PM2.5	Persistence, and an external coupled product (CAM5 or similar)	24 h, 48 h, 72 h
Regime detection	Held-out AUC, hit rate, false-alarm rate	Negative ventilation coefficient	Per episode
Lead time	Median hours from alert to onset	—	Per locked-in episode
Uncertainty	Interval coverage, spread–error correlation	Nominal coverage	All horizons
Counterfactual	Reproduction of an observed emissions perturbation	2020 lockdown, Diwali spikes	Case study
Decision layer	Response latency	—	Separately, and never called forecast latency

HOLD-OUT DISCIPLINE

November 2023 and November 2024 are excluded from every fitting step in the pipeline as written. Those two months are the most severe recent post-monsoon episodes. Any number quoted publicly comes from them.

9 Risk register

RISK	SEVERITY	MITIGATION, AND WHO OWNS IT
ERA5 PBLH is biased in exactly the shallow stable layers that matter most. Known to struggle with nocturnal and heavily-loaded cases.	High	Module B quantifies the bias against radiosonde rather than ignoring it, and the emulator is allowed to learn a correction. Stated openly in the presentation.
λ does not separate the classes on real data.	High	This is why the gate exists in week one. Fallback positioning is already written: coupled forecasting with an explicit feedback diagnostic, no lock-in claim.
Simultaneity bias in the elasticity estimates.	Medium	Documented in section 3.5. Module B replaces OLS with instrumental variables.
Training reference for the emulator cannot be obtained in time.	High	Do not architect around a specific model before confirming data access. Module C decides in week two with a written go/no-go.
CPCB dry mass versus satellite wet AOD are not the same quantity.	Medium	Hygroscopic growth correction is required the moment AOD enters. Module A flags it; Module C implements it.
Scope inflation — the architecture is larger than five people.	High	Module boundaries are drawn to be independently shippable. If a module slips, the others still demo. Cut Module C's spatial resolution before cutting verification.
Presenting the 23 ms latency as forecast speed.	Medium	Removed from all material. Module E owns compliance.

10 What to run first

Steps one to three are one person for about a week and they either prove or kill the central idea. Nothing else should start on the assumption that λ works.

```
cd d:\AZU0\chat\aree-ps26082

# already passing - no keys needed, proves the estimator works
..\venv\Scripts\python.exe scripts\00_selftest_synthetic.py

# 1. confirm CDS access, then pull ERA5 <-- GATES EVERYTHING
set CDSAPI_KEY=<key>
..\venv\Scripts\python.exe scripts\01_fetch_era5.py --dry-run
..\venv\Scripts\python.exe scripts\01_fetch_era5.py

# 2. ground PM2.5
set OPENAQ_API_KEY=<key>
..\venv\Scripts\python.exe scripts\02_fetch_ground_aq.py

# 3. build the panel, estimate lambda, label episodes, validate
..\venv\Scripts\python.exe scripts\03_build_panel.py
..\venv\Scripts\python.exe scripts\04_compute_lambda.py
..\venv\Scripts\python.exe scripts\05_label_episodes.py
..\venv\Scripts\python.exe scripts\06_validate_lambda.py
```

ORDER	TASK	OWNER	GATE
1	Confirm Copernicus CDS access and pull ERA5 blh, ssrd, ssrdc for five winters.	Module A	If PBLH cannot be obtained at hourly cadence, λ as formulated is dead. Know this first.
2	Ground PM2.5 for the same hours.	Module A	—
3	Run the validation gate. Report held-out AUC and lead time.	Module B	Decides whether the lock-in claim is made at all.
4	Settle the emulator training reference.	Module C	Written go/no-go on data availability before any model code.
5	Retune the persistence constants and add the predicted-breach path.	Module D	—
6	Only now: build the emulator.	Module C	—

The honest summary: the architecture is sound, the estimator is written and passes its self-test, and the entire scientific claim now rests on one week of data work that has not yet been done. That is the right place for the risk to sit.